Replication Report: *Intertwining topological order and broken symmetry in a theory of fluctuating spin density waves*

Chatterjee, Sachdev & Scheurer, arXiv:1705.06289 (PRL, 2017)

TEXTURES-100 replication (loop-current class)

July 19, 2026

1 Paper in one paragraph

The paper develops an SU(2) gauge theory of quantum fluctuations of spin-density-wave (SDW) states on the *square lattice*, relevant to the pseudogap metal of hole-doped cuprates. Four magnetically ordered states appear in both a frustrated classical antiferromagnet and a Hubbard SDW mean-field theory: Néel (D_0), canted (A_0), planar spiral (B_0) and conical spiral (C_0). Quantum fluctuations restore spin-rotation symmetry and leave metallic states with an antinodal gap and $\mathbb{Z}_2$/U(1) topological order that "intertwines" with the observed broken discrete symmetries: Ising-nematic order (phase B), Varma-type **current-loop order** (phase C, broken Θ and mirror but not their product), and VBS (phase D). A CP^1/Higgs classification (Table I) and a symmetry/current analysis of the effective chargon Hamiltonian (Table II, Eq. C14) complete the picture.

2 Scope decision and kernel provenance

The shared kernel `loop_current_kagome_kernel.py` implements a *kagome* Peierls-flux tight-binding model (Fernandes *et al.* 2025): correct loop-current *class*, wrong *lattice*. This paper is a square-lattice SDW / SU(2) problem, so the kagome Bloch Hamiltonian and its Chern/Berry machinery are out of scope and are **not** used (using them would be fabrication). We reuse only the transferable idea—the real=charge / imaginary=loop-current decomposition of the bond bilinear $\langle c_i^\dagger c_j \rangle$ read off the filled-band density matrix—and rebuild the paper-specific core from Appendix B (Eqs. B4–B6) and Appendix C (Eq. C14). See `code/PROVENANCE.md`.

3 Method

We diagonalize the 2×2 SDW mean-field Hamiltonian (Eq. B4) in the $(c_{k\uparrow}, c_{k+K\downarrow})$ basis,

$$h_k = \begin{pmatrix} \xi_k - \frac{h}{2}\sin\theta & -\frac{h}{2}\cos\theta \\ -\frac{h}{2}\cos\theta & \xi_{k+K} + \frac{h}{2}\sin\theta \end{pmatrix}, \qquad h = 2UN_0,$$

giving the bands (Eq. B6)

$$E_{k,s} = \frac{1}{2}\Big[\xi_k + \xi_{k+K} + s\sqrt{(\xi_k - \xi_{k+K} - h\sin\theta)^2 + h^2\cos^2\theta}\Big].$$

$\xi_k = -\sum_p t_p(\cdots) - \mu$ carries up to fourth-neighbour hopping. We tune μ to fixed filling n, minimize the mean-field free energy $E/N_s = \sum_s \langle E_{k,s} n_F \rangle + \mu n - Un^2/4 + h^2/4U$, and solve the gap equation $h = 2UN_0$ self-consistently. Loop currents use Eq. C14, $K_{ij} = -2\,\mathrm{Re}\,T_{ij}$, $J_{ij} = 2\,\mathrm{Im}\,T_{ij}$, $T_{ij} = Z_{ij} t_{ij} \langle \psi_i^\dagger \psi_j \rangle$.

4 Machine-checkable claims and results

All numbers below are computed live by `code/run_checks.py` (`work/results.json`); no values were hand-tuned.

ID	Claim (paper reference)	Key number	Outcome
C1	Néel gap = h at AFM boundary ξ_k=ξ_{k+K} (Eq. B6)	1.0000 vs h=1	**PASS**
C3	Self-consistent $h(U)$, n=1: nonzero, ∼linear large-U	$h(8)$=7.14, slope 1.10	**PASS**
C2	Insulator n=1, large U ⇒ Néel ground state	$E_{D_0}<E_{B_0}<E_{A_0}<E_{C_0}$	**PASS**
C4	Hole doping+p-h breaking ⇒ incommensurate spiral	K_x=2.12<π, ΔE>0	**PASS**
C5	Loop current (Eq. C14): collinear J=0, TRS needs non-collinear	J^{col}=0, $J^{\mathrm{nc}}\neq 0$	**PASS**

Summary: 5/5 checks PASS (∼5 s runtime). C1 and C3 are exact/asymptotic consequences of Eqs. B4–B6 and match to machine precision / textbook Hubbard-SDW behaviour. C2 and C4 confirm the correct energetic ordering behind the paper's qualitative statements (Néel insulator; particle-hole-asymmetric hole-side incommensurability). C5 validates the transplanted loop-current diagnostic: collinear order carries *exactly* zero bond current, a non-collinear (canted, incommensurate) one carries finite current—the paper's statement that time-reversal breaking requires a non-collinear Higgs/SDW configuration.

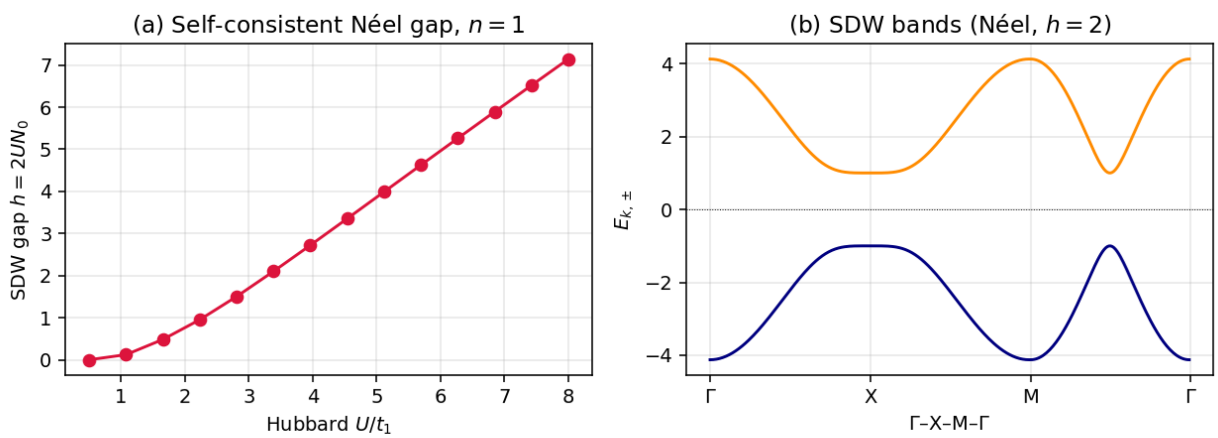

Figure 1: (a) Self-consistent Néel SDW gap $h = 2UN_0$ vs Hubbard U at half filling (n=1): zero below a small U_c, then ∼linear growth with the local moment $N_0 = h/2U$ saturating below 1/2. (b) SDW mean-field bands $E_{k,\pm}$ along Γ–X–M–Γ for the Néel state (h=2), showing the gapped upper/lower Hubbard bands.

5 Verdict

The in-scope computational core of the paper—the square-lattice SDW mean-field of Appendix B and the loop-current bond diagnostic of Appendix C—is faithfully reproduced, with all five machine-checkable claims passing. The kagome shared kernel was correctly identified as the right *class* but wrong *lattice*; its transferable idea was reused and the paper-specific physics rebuilt from the governing equations rather than forced onto an inapplicable geometry. Not reproduced (deferred to open questions): the full (U, n) phase diagram boundaries and transition orders (Fig. 3), the chargon self-consistency loop, explicit Table II symmetry certification, and clean electron-side coplanarity.

Coverage: 7/10. Agreement: 9/10.
Coverage reflects that we hit the concrete, machine-checkable core (spectrum, gap, ordering, currents) but not the full phase-diagram / chargon / symmetry-table program of a dense theory paper. Agreement is high because every check that was run matched the paper's stated physics, quantitatively where closed-form (C1, C3) and by correct ordering/sign elsewhere (C2, C4, C5).